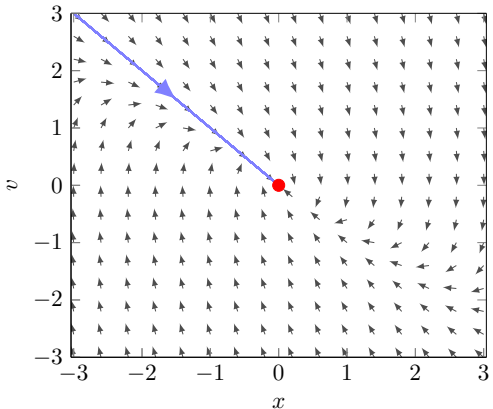
Lecture 5 recap

We learnt about phase space and the first steps of sketching phase portraits.

$$\begin{pmatrix} x \\ v \end{pmatrix} = C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \mathrm{e}^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \quad \boldsymbol{y}' = \mathbf{A} \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= -\mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



Lecture 6

In Lecture 6, we will:

1. Learn how to sketch trajectories when the initial condition does not lie on an eigenvector

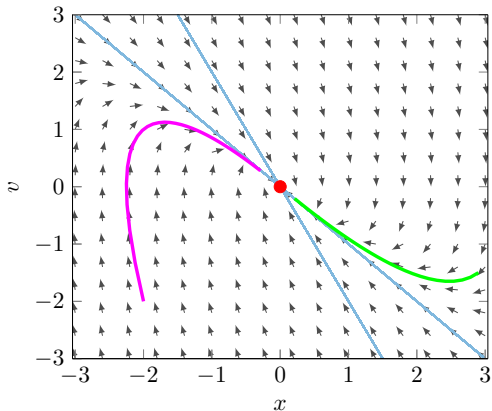
2. Connect the long-term behaviours of solutions to the stabilities of critical points

3. Learn how to solve inhomogeneous systems by diagonalising the matrix $\mathbf{A}$

Sketching phase portraits - curved orbits

What happens if the initial condition is NOT along one of the eigenvectors?

- Linear transformation stretches/compresses/flips AND rotates initial condition.
- Solution is a parametric curve, with arrow pointing in direction of increasing t.
- Solution for large times 'grounded' by eigenvectors.



Sketching phase portraits - curved orbits

Back to Problem 1.2: Recall that

$$\boldsymbol{y}(t) = C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \mathrm{e}^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

- Take an IC

$$\Rightarrow \quad \boldsymbol{y}_0 = C_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad \text{for} \quad C_1, C_2 \neq 0$$

- For large t

$$\boldsymbol{y}(t) \approx C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

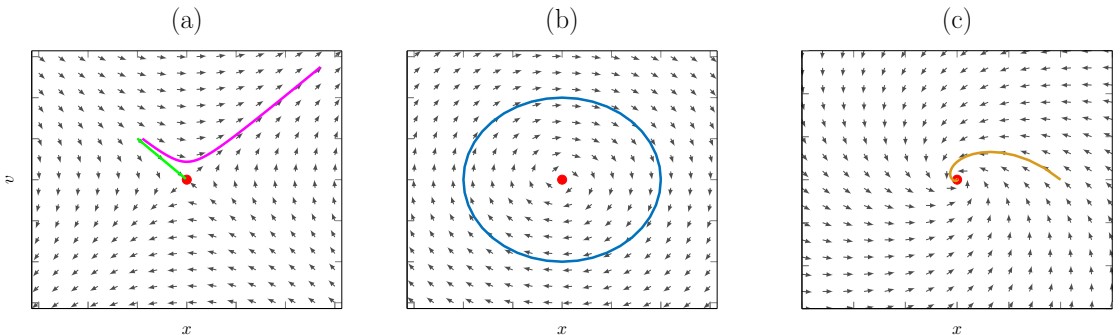
- This tell us that curved trajectories approach critical point along the first eigenvector as $t \to \infty$.

Sketching phase portraits: Stability of critical points

Study of large time behaviour is related to the **stability** of the critical points.

- A critical point is stable if, whenever a solution starts close enough to the critical point, it stays close forever for $t > 0$.
- A critical point is unstable if it is not stable.
  - This means that it is possible for a solution to start arbitrarily close to the critical point and eventually leave the neighbourhood of that point.
- The origin $\boldsymbol{y} = \mathbf{0}$ is stable if the real parts of the eigenvalues are negative or zero. In particular, it is asymptotically stable if the real parts of the eigenvalues are negative.

Exercise 1.13: Which of the following critical points is stable?



Case study 2: Wildlife conservation

- As far back as 15 million years ago, the greater bilby (*Macrotis lagotis*) roamed and inhabited 70 per cent of the Australian mainland. Today, due to predation by animals like dingoes, along with competition for food from rabbits and loss of habitat from urbanisation, this iconic marsupial is listed as a critically endangered species (Blumstein et al (2019) Phil. Trans. R. Soc.B.)
- Understanding the coevolutionary dynamics of predator-prey interactions through mathematical models, specifically **IVPs - inhomogeneous systems**, is important in predicting future populations and developing effective management strategies that allow the population of threatened species like the bilby to recover.





Diagonalisable systems

Revision

If $\mathbf{A}$ be a diagonalisable matrix, then $\mathbf{A} = \mathbf{S}\,\mathbf{D}\,\mathbf{S}^{-1}$ where

- $\mathbf{D}$ is a diagonal matrix whose entries are the eigenvalues
- and the columns of $\mathbf{S}$ contain the eigenvectors.

$$
\begin{aligned}
\boldsymbol{y}' &= \mathbf{A}\boldsymbol{y} + \boldsymbol{b} \\
\boldsymbol{y}' &= \mathbf{S}\mathbf{D}\mathbf{S}^{-1}\boldsymbol{y} + \boldsymbol{b} \\
\mathbf{S}^{-1}\boldsymbol{y}' &= \mathbf{D}\mathbf{S}^{-1}\boldsymbol{y} + \mathbf{S}^{-1}\boldsymbol{b} \\
\boldsymbol{z}' &= \mathbf{D}\boldsymbol{z} + \mathbf{S}^{-1}\boldsymbol{b} \quad \text{where} \quad \boldsymbol{z} = \mathbf{S}^{-1}\boldsymbol{y}.
\end{aligned}
$$

- We have decoupled the system.
- It can be solved as n independent first-order linear ODEs.

Exercise 1.14

Use the method of diagonalisation to solve the following system of ODEs:

$$\begin{aligned} x' - y &= -\mathrm{e}^{-t} \\ y' + 2x + 3y &= \mathrm{e}^{-t} \end{aligned}$$

First we write as a vector equation

$$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix}.$$

The eigenvalues and eigenvectors of the matrix are

$$\lambda_1 = -1 \quad \text{with} \quad \boldsymbol{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \lambda_2 = -2 \quad \text{with} \quad \boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Thus

$$\boldsymbol{S} = \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{S}^{-1} = \begin{pmatrix} -2 & -1 \\ 1 & 1 \end{pmatrix}$$

The diagonalised system is

$$\boldsymbol{z}' = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \boldsymbol{z} + \boldsymbol{S}^{-1} \begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \boldsymbol{z} + \begin{pmatrix} \mathrm{e}^{-t} \\ 0 \end{pmatrix}$$

Solve this decoupled system using integrating factors (exercise) to get

$$\boldsymbol{z} = \begin{pmatrix} (C_1 + t)\,\mathrm{e}^{-t} \\ C_2 \mathrm{e}^{-2t} \end{pmatrix}$$

$$\Rightarrow \quad \boldsymbol{y} = \boldsymbol{S}\boldsymbol{z} = \begin{pmatrix} -(C_1 + t)\,\mathrm{e}^{-t} - C_2 \mathrm{e}^{-2t} \\ (C_1 + t)\,\mathrm{e}^{-t} + 2C_2 \mathrm{e}^{-2t} \end{pmatrix}$$

Solving constant coefficient ODEs using the Matrix Exponential

- Let's have another look at solving the initial value problem

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t)\,, \quad \mathbf{y}(0) = \mathbf{y}_0 \qquad \text{(with } \mathbf{A} \text{ constant)}$$

- Previously: Ansatz $\mathbf{y}(t) = \mathrm{e}^{\lambda t}\mathbf{v}$ with λ eigenvalue and $\mathbf{v}$ eigenvector (a posteriori)
- Why not directly trial $\mathbf{y}(t) = \mathrm{e}^{\mathbf{A}t}\mathbf{y}_0$ instead?

Definition: Matrix Exponential

For an arbitrary square matrix $\mathbf{M}$, the matrix exponential is defined by

$$\mathrm{e}^{\mathbf{M}} = \sum_{n=0}^{\infty} \frac{\mathbf{M}^n}{n!} = I + \mathbf{M} + \frac{1}{2}\mathbf{M}^2 + \cdots$$

Note: This series is always well-defined.

Let's compute the derivative

$$\frac{d}{dt}\mathrm{e}^{\mathbf{A}t} = \frac{d}{dt}\sum_{n=0}^{\infty}\frac{(\mathbf{A}t)^n}{n!} = \sum_{n=0}^{\infty}\frac{\mathbf{A}^n}{n!}\frac{d}{dt}t^n = \sum_{n=1}^{\infty}\frac{\mathbf{A}^n}{(n-1)!}t^{n-1} = \mathbf{A}\,\mathrm{e}^{\mathbf{A}t}$$

This is justified by uniform convergence ($\rightarrow$ Real Analysis)

We furthermore conclude that

$$\mathbf{y}'(t) = \frac{d}{dt}\mathrm{e}^{\mathbf{A}t}\mathbf{y}_0 = \mathbf{A}\mathrm{e}^{\mathbf{A}t}\mathbf{y}_0 = \mathbf{A}\mathbf{y}(t) \quad \text{as well as} \quad \mathbf{y}(0) = \mathrm{e}^{\mathbf{0}}\mathbf{y}_0 = I\mathbf{y}_0 = \mathbf{y}_0\ ,$$

so our ansatz indeed solves the IVP.

Check: If $\mathbf{y}_0 = \mathbf{v}$ is an eigenvector to eigenvalue λ we have $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ and hence

$$\mathbf{y}(t) = \mathrm{e}^{\mathbf{A}t}\mathbf{y}_0 = \sum_{n=0}^{\infty}\frac{t^n}{n!}\mathbf{A}^n\mathbf{v} = \sum_{n=0}^{\infty}\frac{(\lambda t)^n}{n!}\mathbf{v} = \mathrm{e}^{\lambda t}\mathbf{v}$$

Example: Matrix Exponential of a non-diagonalisable matrix

Consider the system of ODEs corresponding to the matrix

$$\mathbf{A} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \qquad \Rightarrow \qquad \text{two-fold degenerate eigenvalue } \lambda$$

This matrix is not diagonalisable and we have

$$\mathbf{A}^2 = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{pmatrix} , \ \dots , \quad \mathbf{A}^n = \begin{pmatrix} \lambda^n & n\lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix} , \ \dots$$

From this we obtain

$$\mathrm{e}^{\mathbf{A}t} = \sum_{n=0}^{\infty} \frac{(\mathbf{A}t)^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \begin{pmatrix} (\lambda t)^n & n(\lambda t)^{n-1} t \\ 0 & (\lambda t)^n \end{pmatrix} = \begin{pmatrix} \mathrm{e}^{\lambda t} & t\mathrm{e}^{\lambda t} \\ 0 & \mathrm{e}^{\lambda t} \end{pmatrix}$$

The matrix exponential knows about both linearly independent solutions $\mathrm{e}^{\lambda t}$ and $t\,\mathrm{e}^{\lambda t}$

Summary of systems of ODEs

1. Higher order ODEs can be written as a single *vector-valued* ODE.
2. Constant coefficient homogeneous: use exponential trial functions.
3. Constant coefficient inhomogeneous: use matrix diagonalisation to decouple the system then use an integrating factor on individual ODEs.

When does it fall apart?

- Matrix is not diagonalisable or coefficients are functions of t:
  learn more about matrix exponentials and path-ordered exponentials!